THE FIRST NEUMANN EIGENSPACE OF A REGULAR TETRAHEDRON: HOT SPOTS AND THE COMPLETE CLASSIFICATION OF GLOBAL EXTREMA

Abstract. We prove the strict hot-spots assertion and determine every global extremum of every first positive Neumann eigenfunction on a solid regular tetrahedron. The eigenspace is three-dimensional, and evaluation at the four vertices identifies it with the zero-sum subspace of $\mathbb{R}^4$. For vertex values $c_1, \dots, c_4$, the global maximum is $\max_i c_i$, attained exactly at the vertices having that value; the analogous statement holds for the minimum. In particular, no nonvertex point of an edge or face is a global extremum, including when several vertex values tie. The proof first establishes a tangent vector-field variational principle and the mean-gradient parametrization. Absolute-value transformations on symmetry chambers then give the monotonicity of the vertex and opposite-edge-pair modes. A local Neumann cone-mean argument makes these monotonicities strict on boundary segments. An ordered nonnegative decomposition into the three types of canonical modes completes the classification. All boundary steps are proved on the solid polyhedron, without assuming a reflection tiling of Euclidean space or an explicit eigenfunction formula.

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1. Introduction and main results

For a bounded connected Lipschitz domain $\Omega \subset \mathbb{R}^3$, write

$$0 = \mu_0(\Omega) < \mu_1(\Omega) \le \mu_2(\Omega) \le \cdots$$

for the Neumann eigenvalues, counted with multiplicity. We use μ_1 for the first positive eigenvalue. Its real eigenspace is

$$E_1(\Omega) = \left\{ u \in H^1(\Omega) : \int_\Omega \nabla u \cdot \nabla \varphi = \mu_1(\Omega) \int_\Omega u\varphi \text{ for every } \varphi \in H^1(\Omega) \right\}. \tag{1.1}$$

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Every member has zero integral. The strict hot-spots assertion says that every nonzero member attains its global extrema only on the boundary. When the eigenvalue has multiplicity greater than one, this is an assertion about every linear combination in the eigenspace.

Let T be the interior of a regular tetrahedron. By translation, rotation and uniform dilation, normalize its vertices by

$$|v_i| = 1, \qquad v_i \cdot v_j = -\frac{1}{3} \quad (i \neq j), \qquad \sum_{i=1}^{4} v_i = 0. \tag{1.2}$$

The theorem below also gives the extrema of its continuous boundary representatives. Their continuity follows from convex-domain Neumann H^2 regularity in dimension three.

Theorem 1.1 (Eigenspace and all global extrema)**.** *Let T be the solid regular tetrahedron just normalized. Then* $\dim E_1(T) = 3$*. There are unique functions $\psi_1, \dots, \psi_4 \in E_1(T)$ such that*

$$\psi_i(v_j) = \delta_{ij} - \frac{1}{4}. \tag{1.3}$$

They satisfy $\sum_i \psi_i = 0$, are permuted by the tetrahedral symmetries, and span $E_1(T)$.

Every $u \in E_1(T)$ has the unique representation

$$u = u_c := \sum_{i=1}^{4} c_i \psi_i, \qquad \sum_{i=1}^{4} c_i = 0, \qquad c_i = u(v_i). \tag{1.4}$$

For $c \neq 0$,

$$\max_{\overline{T}} u_c = \max_{1 \le i \le 4} c_i, \qquad \arg\max_{\overline{T}} u_c = \{v_i : c_i = \max_j c_j\}, \tag{1.5}$$

$$\min_{\overline{T}} u_c = \min_{1 \le i \le 4} c_i, \qquad \arg\min_{\overline{T}} u_c = \{v_i : c_i = \min_j c_j\}. \tag{1.6}$$

In particular, for $c \neq 0$, at every nonvertex point $x \in \overline{T}$,

$$\min_{1 \le i \le 4} u_c(v_i) < u_c(x) < \max_{1 \le i \le 4} u_c(v_i). \tag{1.7}$$

Thus the vertex-evaluation map is an isomorphism

$$E_1(T) \longrightarrow \left\{ c \in \mathbb{R}^4 : \sum_i c_i = 0 \right\}.$$

With the supremum norm on both spaces it is an isometry. A tie of two or three largest values gives exactly two or three maximizing vertices. It does not give a maximizing edge or face. There can be at most three maxima and at most three minima for a nonzero eigenfunction. These statements concern all first eigenfunctions, with no assumption that their coefficient direction is generic. Section 8 gives color-coded TikZ illustrations.

Background. Judge and Mondal proved the hot-spots assertion for Euclidean triangles; their subsequent erratum is part of the reference for that result [4, 5]. Affine reflection alcoves provide another class with explicit spectral structure [2]. The regular solid tetrahedron is not such an alcove: its interior dihedral angle is $\arccos(1/3)$, rather than π/k for an integer $k \geq 2$. Accordingly, an unrestricted reflection of the tetrahedron does not supply a Euclidean tiling from which its first eigenfunctions could simply be read off.

Kennedy and Rohleder developed a vector variational absolute-value method for higher-dimensional lip domains and certain symmetric domains [6]. After suitable orthogonal coordinate changes, the chambers used here have the face-normal pattern needed for that method. We include the particular transformations, equality arguments and Sobolev gluing proofs in full. The flat-face vector identity is the polyhedral Gaffney identity [1, Corollary 6 and Remark 8]; the regularity inputs

are the standard homogeneous Neumann results on convex domains [3, 7]. The additional boundary argument below identifies exact extrema, including all ties, by applying a local cone mean to tangential derivatives. These references explain the ingredients of the proof; the classification is established directly in this paper.

Outline. Section 2 proves the vector variational principle, its equality characterization and the injectivity of the mean gradient. Tetrahedral symmetry identifies the entire first eigenspace in Section 3. We write f_a for the eigenfunction with $\int_T \nabla f_a = a$. Sections 4 and 5 establish the necessary derivative signs for $f_{v_i+v_j}$ and f_{v_i}. Section 6 makes strict interior directional monotonicity strict on every nontrivial segment of the closed chamber, including its edges. Section 7 then identifies the extrema of the canonical modes and, after relabeling the vertices, uses the ordered decomposition

$$a = Av_1 + B(v_1 + v_2) + C(-v_4), \qquad A, B, C \geq 0,$$

to classify every first eigenfunction. This gives a fully analytic proof; no numerical approximation to the eigenvalue or an eigenfunction is used.

2. A variational principle for vector fields on a convex polytope

Throughout this section, $P \subset \mathbb{R}^3$ is a bounded open convex polytope with nonempty interior. Its outward unit normal is constant on each relatively open face. Write

$$\mu = \mu_1(P) > 0, \qquad E = \left\{ u \in H^1(P) : \int_P \nabla u \cdot \nabla \varphi = \mu \int_P u\varphi \quad \text{for every } \varphi \in H^1(P) \right\}.$$

Thus E is the first positive Neumann eigenspace; every $u \in E$ has integral zero. The scalar Poincaré inequality is

$$\int_P |\nabla h|^2 \geq \mu \int_P |h - h_P|^2, \qquad h_P = |P|^{-1} \int_P h. \tag{2.1}$$

If $h_P = 0$, equality holds exactly when $h \in E$, including $h = 0$. Indeed, a nonzero equality case minimizes the mean-zero Rayleigh quotient, and its first variation gives the weak eigenvalue equation.

Lemma 2.1 (Neumann regularity on a convex polytope)**.** *If $g \in L^2(P)$ and $\int_P g = 0$, the mean-zero weak solution of*

$$-\Delta v = g \quad \text{in } P, \qquad \partial_\nu v = 0 \quad \text{on } \partial P$$

belongs to $H^2(P)$. In particular, every element of E belongs to $H^2(P)$.

Proof. Existence and uniqueness on the mean-zero subspace follow from (2.1) and the Lax–Milgram theorem. Since $\int_P g = 0$, subtracting the mean of a test function extends the resulting weak equation to every test function in $H^1(P)$. The asserted regularity is the homogeneous Neumann regularity theorem on bounded convex domains; see [3, Chapter 3] and [7, Section 2.4 and the beginning of Section 4]. Applying it to $g = \mu u$ gives the eigenfunction assertion. Once the solution is in $H^2(P)$, Green's identity and its weak equation show that $\partial_\nu v = 0$ as a Sobolev trace on each relatively open face: test with functions supported near that face and having arbitrary smooth boundary traces there. Thus the weak Neumann condition supplies exactly the face traces used below. □

For every $u \in E$, Sobolev embedding in dimension three gives a continuous representative on $\overline{P}$, so its global extrema exist. Interior elliptic regularity makes u real analytic in P. The normal conditions used below are Sobolev trace conditions on relatively open faces; no pointwise differentiability at edges or vertices is assumed.

Lemma 2.2 (Uniqueness from zero data on a face)**.** *If $w \in E$ has zero trace on a nonempty relatively open subset of a face of P, then $w = 0$ in P. Consequently, if a vector field $Z = (Z_1, Z_2, Z_3)$ has all its components in E and has zero normal trace on every face, then $Z = 0$.*

Proof. Choose a ball crossing that face whose closure meets no other face and whose intersection with the face is contained in the given subset. In the part of the ball inside P, both w and its outward normal derivative have zero trace: the first by assumption and the second by the Neumann eigenvalue equation and H^2 regularity. Extend w by zero to the exterior part of the ball. Its zero Dirichlet trace makes this extension an H^1 function on the ball. Green's identity against compactly supported smooth test functions, using these two zero traces, shows that the extension solves $-\Delta w = \mu w$ distributionally throughout the ball. Interior elliptic regularity and analyticity for the constant-coefficient Helmholtz equation make this extension analytic. It vanishes on the exterior open part of the ball, and hence on the whole ball. Analytic continuation through the connected set P then gives $w = 0$ in P.

For the vector assertion, fix a face normal ν. The scalar function $Z\cdot\nu$ is a linear combination of the Z_j, so it belongs to E and has zero trace on that face. The first assertion gives $Z \cdot \nu = 0$ throughout P. The face normals span $\mathbb{R}^3$: otherwise a nonzero common perpendicular direction would preserve every defining half-space, and P would contain a whole line, contradicting boundedness. Applying this to all face normals gives $Z = 0$. □

Define the space of vector fields with zero normal trace by

$$\mathcal{V}(P) = \left\{ X \in H^1(P;\mathbb{R}^3) : X \cdot \nu = 0 \text{ on every relatively open face} \right\}. \tag{2.2}$$

For $X \in H^1(P;\mathbb{R}^3)$, DX denotes its full Jacobian, with the Euclidean matrix norm. All the vector fields below are real.

Lemma 2.3 (The exact polyhedral vector identity)**.** *For every $X \in \mathcal{V}(P)$,*

$$\int_P |DX|^2 = \int_P \left((\operatorname{div} X)^2 + |\operatorname{curl} X|^2 \right). \tag{2.3}$$

Proof. Apply [1, Corollary 6, equation (9)] with the full normal boundary condition, that is, $\Gamma_n = \partial P$ and $\Gamma_t = \varnothing$ in that reference. Since P is a bounded Lipschitz polyhedron in $\mathbb{R}^3$, [1, Remark 8] identifies the H^1 closure of smooth fields satisfying this condition with the space of H^1 fields having zero normal trace. This is precisely $\mathcal{V}(P)$; edges have surface measure zero and do not impose additional face-trace conditions. The polyhedral equality in the cited corollary is therefore (2.3) for every $X \in \mathcal{V}(P)$. □

In particular, if $u \in E$, then $\nabla u \in \mathcal{V}(P)$ by Lemma 2.1. Its curl vanishes and $\Delta u = -\mu u$, so (2.3) and the weak eigenvalue equation tested with u give

$$\int_P |D^2 u|^2 = \int_P (\Delta u)^2 = \mu^2 \int_P u^2 = \mu \int_P |\nabla u|^2. \tag{2.4}$$

Proposition 2.4 (The lowest vector energy and its equality space)**.** *Every $X \in \mathcal{V}(P)$ satisfies*

$$\int_P |DX|^2 \geq \mu \int_P |X|^2. \tag{2.5}$$

Equality holds if and only if $X = \nabla u$ for some $u \in E$. Thus the equality space is exactly ∇E.

Proof. The divergence theorem for H^1 vector fields and their traces gives $\int_P \operatorname{div} X = 0$. Lemma 2.1 therefore supplies a unique mean-zero $v \in H^2(P)$ solving

$$\Delta v = \operatorname{div} X \quad \text{in } P, \qquad \partial_\nu v = 0 \quad \text{on } \partial P.$$

Its weak formulation, using $X \cdot \nu = 0$, is

$$\int_P \nabla v \cdot \nabla \varphi = -\int_P (\operatorname{div} X)\varphi = \int_P X \cdot \nabla \varphi \qquad (\varphi \in H^1(P)).$$

Set $Y = X - \nabla v$. Then $Y \in \mathcal{V}(P)$, $\operatorname{div} Y = 0$, and

$$\int_P Y \cdot \nabla\varphi = 0 \quad \text{for all } \varphi \in H^1(P). \tag{2.6}$$

In particular, Y is L^2 orthogonal to ∇v and $\int_P Y_j = 0$ for $j = 1, 2, 3$, by taking $\varphi = v$ and $\varphi = x_j$, respectively. Consequently

$$\int_P |X|^2 = \int_P |\nabla v|^2 + \int_P |Y|^2. \tag{2.7}$$

Since $\operatorname{curl} \nabla v = 0$, two applications of (2.3) give the exact energy decomposition

$$\begin{aligned} \int_P |DX|^2 &= \int_P (\Delta v)^2 + \int_P |\operatorname{curl} Y|^2 \\ &= \int_P (\Delta v)^2 + \int_P |DY|^2. \end{aligned} \tag{2.8}$$

Since $v_P = 0$ and $\partial_\nu v = 0$, Green's identity, Cauchy–Schwarz and (2.1) give

$$\int_P |\nabla v|^2 = -\int_P v\Delta v \le \|v\|_{L^2}\|\Delta v\|_{L^2} \le \mu^{-1/2}\|\nabla v\|_{L^2}\|\Delta v\|_{L^2}.$$

If $v \neq 0$, its zero mean implies $\|\nabla v\|_{L^2} > 0$. Dividing the displayed inequality by this norm and squaring gives

$$\int_P (\Delta v)^2 \ge \mu \int_P |\nabla v|^2. \tag{2.9}$$

Equality forces equality in the Poincaré step, hence $v \in E$. Conversely, $v \in E$ gives equality by its eigenvalue equation. The zero function is included in this characterization. Applying (2.1) separately to the three mean-zero components of Y gives

$$\int_P |DY|^2 \ge \mu \int_P |Y|^2. \tag{2.10}$$

Equations (2.7)–(2.10) prove (2.5).

Suppose equality holds. Both nonnegative deficits in (2.9) and (2.10) vanish. Hence $v \in E$ and every component Y_j belongs to E. Since Y has zero normal trace, the vector assertion of Lemma 2.2 gives $Y = 0$. Thus $X = \nabla v$ with $v \in E$.

Conversely, if $u \in E$, then $\nabla u \in \mathcal{V}(P)$ and (2.4) shows that this gradient attains equality. □

Corollary 2.5 (Rigidity of a modification preserving the energy)**.** *Let $0 \neq u \in E$. If $X \in \mathcal{V}(P)$ satisfies*

$$\int_P |X|^2 = \int_P |\nabla u|^2, \qquad \int_P |DX|^2 = \int_P |D^2 u|^2,$$

then $X = \nabla w$ for some $w \in E$.

Proof. The hypotheses and (2.4) make the Rayleigh quotient of X equal to μ; Proposition 2.4 characterizes its equality cases. □

Lemma 2.6 (Injectivity and covariance of the mean gradient)**.** *The linear map*

$$J : E \longrightarrow \mathbb{R}^3, \qquad J(u) = \int_P \nabla u, \tag{2.11}$$

is injective. In particular, $\dim E \le 3$. *It also satisfies*

$$J(u) = \mu \int_P x\, u(x)\, dx. \tag{2.12}$$

If an orthogonal map R preserves P, then

$$J(u \circ R^{-1}) = RJ(u). \tag{2.13}$$

Proof. Suppose $J(u) = 0$. Each derivative $\partial_j u$ belongs to $H^1(P)$ and has mean zero. By (2.4),

$$0 = \int_P |D^2 u|^2 - \mu \int_P |\nabla u|^2 = \sum_{j=1}^{3} \left(\int_P |\nabla \partial_j u|^2 - \mu \int_P |\partial_j u|^2 \right).$$

Every summand is nonnegative by (2.1), so each vanishes and $\partial_j u \in E$ for all j. The weak Neumann conditions for these derivatives follow from this equality characterization; they were not assumed beforehand. Now $\nabla u \in \mathcal{V}(P)$ and its components lie in E, so the vector assertion of Lemma 2.2 gives $\nabla u = 0$. Since $u_P = 0$, this proves $u = 0$ and injectivity.

Taking $\varphi = x_j$ in the weak eigenvalue equation gives $\int_P \partial_j u = \mu \int_P x_j u$, which proves (2.12). If R is orthogonal and preserves P, change of variables in the weak equation first shows that $u \circ R^{-1} \in E$. The chain rule then gives

$$\nabla(u \circ R^{-1})(x) = R \nabla u(R^{-1} x),$$

and integration, using $|\det R| = 1$, proves (2.13). □

3. The first eigenspace and tetrahedral symmetry

We now specialize to the regular tetrahedron T, normalized as in (1.2): its centroid is the origin and its vertices satisfy

$$|v_i| = 1, \qquad v_i \cdot v_j = -\frac{1}{3} \ (i \neq j), \qquad \sum_{i=1}^{4} v_i = 0. \tag{3.1}$$

The Gram matrix of the vertices is $\frac{4}{3} I_4 - \frac{1}{3} \mathbf{1}\mathbf{1}^{\mathsf{T}}$, where $\mathbf{1} = (1,1,1,1)^{\mathsf{T}}$. It has rank three and kernel $\mathbb{R}\mathbf{1}$. Thus the vertices span $\mathbb{R}^3$, and their only linear relation is $\sum_i v_i = 0$, up to a scalar multiple. Every permutation σ of the vertices therefore defines a unique linear map $g_\sigma v_i = v_{\sigma(i)}$. The scalar products in (3.1) show that this map is orthogonal, and it preserves T because it permutes its vertices.

Put $\mu = \mu_1(T)$ and $E = E_1(T)$. The barycentric coordinates $\lambda_i(x)$ are characterized by

$$x = \sum_{i=1}^{4} \lambda_i(x) v_i, \qquad \sum_{i=1}^{4} \lambda_i(x) = 1, \qquad \lambda_i(x) = \frac{1 + 3x \cdot v_i}{4}. \tag{3.2}$$

Indeed, taking the scalar product of the barycentric expansion with v_i gives $x \cdot v_i = (4\lambda_i - 1)/3$. The domain is $T = \{\lambda_i > 0 \text{ for all } i\}$. For a constant vector b, write $D_b u = b \cdot \nabla u$.

Lemma 3.1 (Canonical parametrization of the first eigenspace)**.** *The map*

$$J : E \longrightarrow \mathbb{R}^3, \qquad J(u) = \int_T \nabla u$$

is an isomorphism. In particular, $\dim E = 3$. *For $a \in \mathbb{R}^3$, let $f_a = J^{-1}(a)$. Then this assignment is linear, and for every orthogonal symmetry g of T,*

$$f_a(gx) = f_{g^{-1}a}(x) \qquad (x \in \overline{T}). \tag{3.3}$$

Moreover,

$$\mu \int_T x f_a(x)\, dx = a. \tag{3.4}$$

Proof. By Lemma 2.6, J is injective, and its covariance (2.13) makes $J(E)$ invariant under every orthogonal symmetry of T. Since $E \neq \{0\}$, injectivity also gives $J(E) \neq \{0\}$.

Here is an elementary verification that the resulting action on $\mathbb{R}^3$ is irreducible. Let $W \neq \{0\}$ be invariant and choose $0 \neq w \in W$. Some pair satisfies $w \cdot (v_i - v_j) \neq 0$, because the vertex differences span $\mathbb{R}^3$; indeed, $v_i = \frac{1}{4} \sum_{j=1}^{4} (v_i - v_j)$. Set $d = v_i - v_j$. The reflection in $d^\perp$ is

$$R_{ij} x = x - 2 \frac{x \cdot d}{|d|^2} d.$$

Using (3.1), this reflection interchanges v_i, v_j and fixes each of the other two vertices, so it is a symmetry of T. Hence

$$w - R_{ij}w = 2\frac{w \cdot d}{|d|^2}d \in W \setminus \{0\},$$

which puts $v_i - v_j$ in W. The vertex-permutation symmetries constructed above send this difference to every other vertex difference. Invariance of W and the spanning property therefore give $W = \mathbb{R}^3$. Applied to $W = J(E)$, this proves surjectivity of J, and hence $\dim E = 3$.

The inverse $a \mapsto f_a$ is linear. Applying (2.13) with $R = g^{-1}$ gives $J(f_a \circ g) = g^{-1}a$, so injectivity yields (3.3) as an identity in E. The continuous representatives supplied by Lemma 2.1 extend this identity to $\overline{T}$. Finally, (2.12) applied to f_a gives (3.4). □

4. The mode directed between opposite edge midpoints

Retain the notation $E = E_1(T)$, $\mu = \mu_1(T)$ and $f_a = J^{-1}(a)$ from Section 3. All derivatives below are weak derivatives when Sobolev spaces are being used, and their smooth interior representatives when pointwise inequalities are stated.

Lemma 4.1 (The midpoint mode)**.** *Choose orthogonal coordinates (x, y, z) in which*

$$\begin{aligned} v_1 &= (a, 0, b), \quad & v_2 &= (-a, 0, b), \\ v_3 &= (0, a, -b), \quad & v_4 &= (0, -a, -b), \end{aligned} \qquad a = \sqrt{\frac{2}{3}}, \quad b = \frac{1}{\sqrt{3}}. \tag{4.1}$$

Then $h = f_{v_1+v_2}$ satisfies

$$h_z > 0, \qquad \operatorname{sgn}(x)h_x \geq 0, \qquad -\operatorname{sgn}(y)h_y \geq 0 \quad \text{in } T. \tag{4.2}$$

The last two inequalities are strict when $x \neq 0$ and $y \neq 0$, respectively. They are understood together with $h_x = 0$ on $x = 0$ and $h_y = 0$ on $y = 0$. Consequently, for any distinct indices i, j,

$$D_{v_i} f_{v_i+v_j}(p) > 0 \quad \text{if } p \in T \text{ and } \lambda_i(p) \geq \lambda_j(p). \tag{4.3}$$

Proof. The vectors

$$e_x = \frac{v_1 - v_2}{2a}, \qquad e_y = \frac{v_3 - v_4}{2a}, \qquad e_z = \frac{v_1 + v_2}{2b}$$

are orthonormal by (3.1) and give the coordinates (4.1). In these coordinates, the barycentric formula (3.2) reads

$$\begin{aligned} \lambda_1 &= \frac{z+b}{4b} + \frac{x}{2a}, \qquad & \lambda_2 &= \frac{z+b}{4b} - \frac{x}{2a}, \\ \lambda_3 &= \frac{b-z}{4b} + \frac{y}{2a}, \qquad & \lambda_4 &= \frac{b-z}{4b} - \frac{y}{2a}. \end{aligned}$$

Since $2b/a = \sqrt{2}$, their positivity gives

$$T = \{(x, y, z) : -b + \sqrt{2}|x| < z < b - \sqrt{2}|y|\}. \tag{4.4}$$

The coordinate reflections in x and y preserve T and fix $v_1 + v_2 = 2be_z$. Thus (3.3) makes h even in each of x, y.

Define, almost everywhere in T,

$$X = (\operatorname{sgn}(x)|h_x|, -\operatorname{sgn}(y)|h_y|, |h_z|). \tag{4.5}$$

Here and below the value assigned to $\operatorname{sgn}(0)$ is irrelevant to Sobolev equivalence classes. Lemma 2.1 gives $h \in H^2(T)$, so its derivatives belong to $H^1(T)$. Differentiating the reflection symmetries weakly makes h_x odd in x and h_y odd in y. Since an H^1 function has a single trace across an interior plane, each corresponding mirror trace equals its negative and is therefore zero. The trace commutes with the Lipschitz map $s \mapsto |s|$, so absolute values preserve these zero traces. Thus the two pieces defining $\operatorname{sgn}(x)|h_x|$ have matching, zero traces across $x = 0$; the analogous statement holds for the second component across $y = 0$. The Sobolev gluing criterion consequently gives

$X \in H^1(T;\mathbb{R}^3)$. Explicitly, the distributional derivative of a function given by H^1 pieces has no surface jump term when their traces agree, which is the criterion being used here.

On the lower and upper physical faces of (4.4), respectively, the Neumann conditions are

$$\sqrt{2}\,\mathrm{sgn}(x)h_x - h_z = 0, \qquad \sqrt{2}\,\mathrm{sgn}(y)h_y + h_z = 0. \tag{4.6}$$

The coordinate x has fixed sign on each relatively open lower face, and y has fixed sign on each relatively open upper face. Taking absolute values of the Sobolev traces in these relations shows that the field (4.5) is tangent to all four faces. For example, its normal component on a lower face, using the unnormalized normal $(\sqrt{2}\,\mathrm{sgn}(x), 0, -1)$, is $\sqrt{2}|h_x| - |h_z| = 0$. The upper faces give $-\sqrt{2}|h_y| + |h_z| = 0$. Thus $X \in \mathcal{V}(T)$.

The Sobolev chain rule for absolute values gives

$$|X| = |\nabla h|, \qquad |DX|^2 = |D^2 h|^2 \quad \text{almost everywhere in } T. \tag{4.7}$$

For the second equality one uses $D|g| = \mathrm{sgn}(g)Dg$ almost everywhere and the fact that $Dg = 0$ almost everywhere on $\{g = 0\}$ for $g \in H^1$. The fixed spatial signs in (4.5) introduce no additional weak derivative because the traces already match.

Since $J(h) = 2be_z \neq 0$, the eigenfunction h is nonzero. Corollary 2.5, applied using (4.7), now supplies $\widetilde{h} \in E$ with $X = \nabla\widetilde{h}$. The first component of X is odd in x, and the second is odd in y; reflection invariance of T therefore makes their integrals zero. Consequently

$$J(\widetilde{h}) = \int_T X = \left(\int_T |h_z|\right) e_z = c_0 J(h),$$
$$c_0 = \frac{1}{2b}\int_T |h_z| > 0,$$

where positivity follows from $\int_T h_z = 2b > 0$. Injectivity of J gives $\widetilde{h} = c_0 h$. The first identity in (4.7) gives $c_0^2 = 1$, since $\nabla h \not\equiv 0$. Consequently $c_0 = 1$, and $X = \nabla h$. This proves all the weak inequalities in (4.2).

In the interior, h_z is smooth, nonnegative, and solves $-\Delta h_z = \mu h_z$. Its integral is $2b > 0$. Since $\Delta h_z = -\mu h_z \leq 0$, the strong minimum principle therefore gives $h_z > 0$ throughout the connected domain T. The other inequalities hold pointwise in the interior by continuity. We also record the strict transverse signs on the convex quarter chamber

$$Q = T \cap \{x > 0, y > 0\}.$$

At a point of a relatively open physical face, choose a ball meeting no other face. The even extension of h across its face plane satisfies the weak Helmholtz equation on that ball: testing the original Neumann equation with the sum of a test function and its reflection gives the equation for the extension. Interior elliptic regularity therefore makes the extension smooth.

If h_x were identically zero on Q, write $h = H(y,z)$ there. This representation is valid on the whole projection because every x-fiber of Q is an interval. Its (y,z)-projection is

$$D_x = \{(y,z) : y > 0,\ -b < z < b - \sqrt{2}y\}.$$

For every $(y,z) \in D_x$, the point with $x = (z+b)/\sqrt{2}$ lies on a relatively open lower physical face. The smooth extension just established lets us evaluate the first relation of (4.6) at this endpoint using the limits of the derivatives along the x-fiber. Since $h_x = 0$ and $h_z = H_z(y,z)$ along that fiber, it gives $H_z(y,z) = 0$. This would make $h_z = 0$ on Q, a contradiction. Likewise, if $h_y \equiv 0$ on Q, write $h = K(x,z)$. Its (x,z)-projection is

$$D_y = \{(x,z) : x > 0,\ -b + \sqrt{2}x < z < b\}.$$

The endpoint $y = (b-z)/\sqrt{2}$ belongs to a relatively open upper face, where the second relation of (4.6) gives $K_z = 0$, again a contradiction. Thus h_x and $-h_y$ are nonzero on Q. The already established weak signs and interior continuity make them nonnegative Helmholtz solutions on $T \cap$

$\{x > 0\}$ and $T \cap \{y > 0\}$, respectively. Each half-domain is convex and contains Q, so the strong minimum principle gives strict positivity throughout it. This includes the internal mirrors $y = 0$ and $x = 0$, respectively. Reflection parity supplies the signs on the negative halves and the zero values of h_x on $x = 0$ and h_y on $y = 0$.

Finally, $x = a(\lambda_1 - \lambda_2)$ in (4.1). At every interior point with $\lambda_1 \geq \lambda_2$, one has $h_x \geq 0$, including $h_x = 0$ when $x = 0$. Hence

$$D_{v_1} h = a h_x + b h_z > 0.$$

Relabeling the vertices proves (4.3) for every pair $i \neq j$. □

5. The vertex modes and their symmetry chambers

Lemma 5.1 (Signs for a vertex mode)**.** *Choose orthogonal coordinates (r, t, z) such that*

$$\begin{aligned} v_4 &= (0, 0, 1), \\ v_1 &= \left(\frac{R}{2}, \frac{\sqrt{3}R}{2}, -\frac{1}{3}\right), \\ v_2 &= \left(\frac{R}{2}, -\frac{\sqrt{3}R}{2}, -\frac{1}{3}\right), \qquad R = \frac{2\sqrt{2}}{3}. \\ v_3 &= \left(-R, 0, -\frac{1}{3}\right), \end{aligned} \tag{5.1}$$

Let $f = f_{v_4}$ and

$$C = \{p \in T : \lambda_1(p) > \lambda_2(p) > \lambda_3(p)\}.$$

Then

$$f_z > 0 \quad \text{throughout } T, \qquad f_r < 0, \quad f_t < 0 \quad \text{on } C. \tag{5.2}$$

The nonstrict versions of the last two inequalities hold at interior points of T lying on the walls of C.

Proof. The four displayed vectors satisfy (3.1), so the Gram-matrix argument of Section 3 justifies the coordinate choice. Let $\mathcal{G}$ consist of the six orthogonal symmetries induced by permutations of v_1, v_2, v_3 and fixing v_4. By (3.3), $f = f_{v_4}$ is invariant under $\mathcal{G}$.

We first verify the chamber geometry and all the relevant normals. For a point $p = (r, t, z) = \sum_{i=1}^{4} \lambda_i v_i$, one has

$$\begin{aligned} r &= \frac{R}{2}(\lambda_1 + \lambda_2 - 2\lambda_3), \\ t &= \frac{\sqrt{3}R}{2}(\lambda_1 - \lambda_2), \\ \sqrt{3}r - t &= \sqrt{3}R(\lambda_2 - \lambda_3), \\ z &= \frac{4\lambda_4 - 1}{3}. \end{aligned} \tag{5.3}$$

In particular, the two strict ordering inequalities defining C are exactly $0 < t < \sqrt{3}r$; both ordering inequalities are needed. Put

$$c = \frac{\sqrt{2}}{4} > 0.$$

The formula for r also gives

$$c(1 - z) - r = \sqrt{2}\lambda_3. \tag{5.4}$$

It follows that

$$C = \{(r, t, z) : z > -1/3,\ 0 < t < \sqrt{3}r,\ r + cz < c\}. \tag{5.5}$$

Here $r > 0$ and $r < c(1-z)$ also give $z < 1$. Conversely, (5.3) and (5.4) show that a point satisfying (5.5) has $\lambda_4 > 0$ and $\lambda_1 > \lambda_2 > \lambda_3 > 0$, so it belongs to the stated chamber in T.

Write $s_{12}, s_{23} \in \mathcal{G}$ for the transpositions of the indicated vertices. Their mirror planes are $t = 0$ and $t = \sqrt{3}r$, respectively. Invariance of f gives zero normal trace of ∇f on these mirrors, by the odd-derivative trace argument in Lemma 4.1. The other two chamber faces, $z = -1/3$ and $r + cz = c$, are the physical faces $\lambda_4 = 0$ and $\lambda_3 = 0$, where tangency is the Neumann condition. Define on C

$$X_C = (-|f_r|, -|f_t|, |f_z|). \tag{5.6}$$

Lemma 2.1 gives $f \in H^2(T)$, and the absolute-value chain rule used in Lemma 4.1 gives $X_C \in H^1(C; \mathbb{R}^3)$. Absolute values commute with the Sobolev traces, as in that lemma. The following table lists unnormalized normals and the trace relations. Each relation in the final column follows from the preceding column; the final column is precisely zero normal trace for X_C.

face	normal	relation for ∇f	relation for X_C
$t = 0$	$(0, -1, 0)$	$f_t = 0$	$\lvert f_t\rvert = 0$
$t = \sqrt{3}r$	$(-\sqrt{3}, 1, 0)$	$-\sqrt{3}f_r + f_t = 0$	$\sqrt{3}\lvert f_r\rvert - \lvert f_t\rvert = 0$
$z = -1/3$	$(0, 0, -1)$	$f_z = 0$	$-\lvert f_z\rvert = 0$
$r + cz = c$	$(1, 0, c)$	$f_r + cf_z = 0$	$-\lvert f_r\rvert + c\lvert f_z\rvert = 0$

In the last row the positivity of c is used. Thus X_C is tangent to all four chamber faces, including both mirrors.

We next explain the extension across the mirrors. Up to their boundaries, the six sets gC, $g \in \mathcal{G}$, partition T; they correspond to the six possible strict orderings of $\lambda_1, \lambda_2, \lambda_3$. Define

$$X(gp) = gX_C(p), \qquad p \in C, \quad g \in \mathcal{G}. \tag{5.7}$$

This defines X almost everywhere. Every g is orthogonal and fixes $e_z = v_4$. To verify the matching traces, fix $s \in \{s_{12}, s_{23}\}$. For almost every point p on its mirror face in $\overline{C}$, the adjacent prescriptions from gC and gsC at gp have traces $gX_C(p)$ and $gsX_C(p)$. The normal table gives $sX_C(p) = X_C(p)$, so these traces agree. Intersections of more than two chamber walls have surface measure zero in the interfaces and do not affect this trace argument.

The matching-trace gluing criterion used in Lemma 4.1 now gives $X \in H^1(T; \mathbb{R}^3)$. On each physical face, orthogonal transport of the tangency relations in the table gives $X \cdot \nu = 0$. Consequently $X \in \mathcal{V}(T)$.

The absolute-value chain rule gives $|X_C| = |\nabla f|$ and $|DX_C|^2 = |D^2 f|^2$ almost everywhere on C. Invariance of f gives $\nabla f(gp) = g\nabla f(p)$. Differentiating within each chamber gives, almost everywhere,

$$DX(gp) = g\, DX_C(p) g^{-1}, \qquad D^2 f(gp) = g\, D^2 f(p) g^{-1}.$$

Orthogonal transport preserves both vector and Frobenius norms, so the extension satisfies

$$|X| = |\nabla f|, \qquad |DX|^2 = |D^2 f|^2 \quad \text{almost everywhere on } T. \tag{5.8}$$

Since $J(f) = v_4 \neq 0$, Corollary 2.5 applies to (5.8) and supplies $\widetilde{f} \in E_1(T)$ with $X = \nabla \widetilde{f}$. The construction gives $X(gp) = gX(p)$ for every $g \in \mathcal{G}$, almost everywhere, so a change of variables makes $J(\widetilde{f}) = \int_T X$ fixed by $\mathcal{G}$. The two generating mirror planes intersect in the z-axis; thus the fixed-vector space is exactly $\mathbb{R}v_4$. Moreover, $ge_z = e_z$ and invariance of f give $X_z = |f_z|$ almost everywhere on T. Consequently

$$J(\widetilde{f}) = \left(\int_T |f_z| \right) v_4 = c_0 J(f),$$
$$c_0 = \int_T |f_z| > 0,$$

where positivity follows from $\int_T f_z = J(f)\cdot v_4 = 1$. Injectivity of J gives $\widetilde{f} = c_0 f$. Equality of the norms in (5.8) gives $c_0^2 = 1$, so $c_0 = 1$. Therefore $X = \nabla f$. Interior continuity makes the resulting weak signs pointwise: $f_z \geq 0$ on T and $f_r, f_t \leq 0$ on C.

The function f_z is smooth in the interior, satisfies $-\Delta f_z = \mu f_z$, and has integral one. The strong minimum principle applied to the nonnegative superharmonic function f_z therefore gives $f_z > 0$ throughout T. To see that both transverse signs are strict in C, first suppose $f_r \equiv 0$ there. The (t,z)-projection of C is

$$D_r = \{(t,z) : z > -1/3,\ 0 < t < \sqrt{3}c(1-z)\},$$

and the fiber over $(t,z) \in D_r$ is $t/\sqrt{3} < r < c(1-z)$. Independence along these interval fibers gives $f = F(t,z)$. Their endpoints at $r = c(1-z)$ lie on the relatively open physical face $\lambda_3 = 0$. The local even Neumann reflection proved in Lemma 4.1 makes f smooth across this face. The Neumann relation $f_r + cf_z = 0$ at the endpoint, evaluated using the smooth limit along the fiber, therefore gives $cF_z = 0$ for every $(t,z) \in D_r$. This would imply $f_z = 0$ throughout C, contradicting $f_z > 0$.

Next, suppose $f_t \equiv 0$ on C. Its (r,z)-projection is

$$D_t = \{(r,z) : z > -1/3,\ 0 < r < c(1-z)\},$$

with fibers $0 < t < \sqrt{3}r$, so $f = H(r,z)$ there. The endpoints $t = \sqrt{3}r$ lie on the internal mirror inside T, where interior smoothness and $-\sqrt{3}f_r + f_t = 0$ give $H_r = 0$ on all of D_t. Hence $f_r \equiv 0$ on C, the case just excluded. Thus $-f_r$ and $-f_t$ are nonzero nonnegative Helmholtz solutions on the connected chamber C. The strong minimum principle makes both strictly positive there. Interior continuity gives the nonstrict inequalities on its walls lying inside T. □

Corollary 5.2 (Directional inequalities for vertex modes)**.** *For every index j,*

$$D_{v_j} f_{v_j}(p) > 0 \qquad (p \in T). \tag{5.9}$$

For distinct indices i, j and an interior point $p \in T$, if

$$\lambda_i(p) \geq \lambda_k(p) \qquad \text{for every } k \neq j, \tag{5.10}$$

then

$$D_{v_i} f_{v_j}(p) \leq -\frac{1}{3} D_{v_j} f_{v_j}(p) < 0. \tag{5.11}$$

In particular, under (5.10), $D_{v_i} f_{-v_j}(p) > 0$.

Proof. The first assertion is $f_z > 0$ in the coordinates of Lemma 5.1, followed by relabeling. For the second, first take $j = 4$ and $i = 1$ in (5.1). If $\lambda_1 \geq \lambda_2 \geq \lambda_3$, the point lies in the closure of C within T. The signs in (5.2) yield

$$\begin{aligned} D_{v_1} f_{v_4} &= \frac{R}{2} f_r + \frac{\sqrt{3}R}{2} f_t - \frac{1}{3} f_z \\ &\leq -\frac{1}{3} f_z = -\frac{1}{3} D_{v_4} f_{v_4} < 0. \end{aligned}$$

If instead $\lambda_1 \geq \lambda_3 \geq \lambda_2$, use the symmetry interchanging v_2, v_3, which fixes both v_1, v_4. By (3.3) and the chain rule, f_{v_4} and the directional derivative $D_{v_1} f_{v_4}$ are invariant under that symmetry, so the same inequality holds. The two weak orderings include all ties and together are exactly the condition that λ_1 be maximal among $\lambda_1, \lambda_2, \lambda_3$. Relabeling proves (5.11) for all $i \neq j$. Finally $f_{-v_j} = -f_{v_j}$ by linearity. □

6. Strict monotonicity on closed polyhedral segments

The preceding derivative inequalities were first obtained in the interior. This section supplies the boundary step needed to exclude additional extrema on edges and faces. The argument applies to a convex polyhedral chamber, with homogeneous Neumann conditions on both its physical faces and its symmetry mirrors.

Lemma 6.1 (A local Neumann solution is continuous at a corner)**.** *Let $P \subset \mathbb{R}^3$ be a bounded convex polytope, $q \in \overline{P}$, and let $\rho > 0$ be sufficiently small that $B_\rho(q)$ meets no face plane not containing q. Fix $\mu > 0$. Suppose $g \in H^1(P \cap B_\rho(q))$ satisfies $-\Delta g = \mu g$ locally, with weak homogeneous Neumann data on the incident faces. More explicitly,*

$$\int_{P\cap B_\rho(q)} \nabla g \cdot \nabla \eta = \mu \int_{P\cap B_\rho(q)} g\eta$$

for all H^1 test functions supported away from the spherical boundary. Then g has a continuous representative on $\overline{P} \cap B_{\rho/2}(q)$.

Proof. Choose a smooth radial cutoff $\chi(x) = \chi_0(|x - q|)$ which equals one on $B_{\rho/2}(q)$ and has compact support in $B_\rho(q)$. For each incident face with constant normal n, its plane contains q, so

$$\partial_n \chi(x) = 0 \qquad \text{on that face.}$$

The product $w = \chi g$, extended by zero to the rest of P, belongs to $H^1(P)$. Products of g or its derivatives with χ or derivatives of χ are likewise extended by zero outside the cutoff support. In particular, $g\nabla\chi \in H^1(P;\mathbb{R}^3)$. Its normal trace vanishes on incident faces by the displayed identity, and its support avoids every other face. Integration by parts therefore gives, for every $\eta \in H^1(P)$,

$$\int_P g\nabla\chi \cdot \nabla\eta = -\int_P (\nabla g \cdot \nabla\chi + g\Delta\chi)\eta.$$

Using the local weak equation with the test function $\chi\eta$ and expanding ∇w now yields

$$\int_P \nabla w \cdot \nabla\eta = \int_P F\eta, \qquad F = \mu\chi g - 2\nabla\chi \cdot \nabla g - g\Delta\chi \in L^2(P).$$

Testing this equation with 1 gives $\int_P F = 0$. Lemma 2.1, applied to $w - w_P$, yields $w \in H^2(P)$. The Sobolev embedding $H^2(P) \hookrightarrow C^{0,1/2}(\overline{P})$ in dimension three proves the assertion where $\chi = 1$. □

Lemma 6.2 (The local Neumann cone mean)**.** *In the setting of Lemma 6.1, let $0 < R < \rho/2$. Write $V_R = P \cap B_R(q)$. Then*

$$\frac{1}{|V_R|}\int_{V_R} g = A_0(\sqrt{\mu}R)g(q), \qquad A_0(t) = \frac{3(\sin t - t\cos t)}{t^3}, \quad A_0(0) = 1. \tag{6.1}$$

In particular, if $g > 0$ in V_R and $0 < \sqrt{\mu}R < \pi/2$, then $g(q) > 0$.

Proof. Write $P = \{\ell_j > 0 : 1 \le j \le N\}$ using affine functions for its face inequalities. If $\ell_j(q) > 0$, the corresponding face plane avoids $B_\rho(q)$, so ℓ_j stays positive throughout that ball. The remaining inequalities are homogeneous after translation by q. Thus

$$K_q = \{y \in \mathbb{R}^3 : \nabla\ell_j \cdot y > 0 \text{ whenever } \ell_j(q) = 0\},$$
$$P \cap B_\rho(q) = q + (K_q \cap B_\rho(0)).$$

The spherical link $\Sigma = K_q \cap S^2$ is independent of radius and has positive area. Define

$$m(r) = \frac{1}{|\Sigma|}\int_\Sigma g(q + r\omega)\, d\omega.$$

Polar coordinates on compact radial intervals and $g \in H^1$ give $m \in H^1_{\mathrm{loc}}(0, R)$, with

$$m'(r) = \frac{1}{|\Sigma|}\int_\Sigma \nabla g(q + r\omega) \cdot \omega\, d\omega \quad \text{for almost every } r.$$

For $\phi \in C_c^\infty(0,R)$, use $\eta(x) = \phi(|x-q|)$ in the local weak equation to obtain

$$\int_0^R r^2 m'(r)\phi'(r)\,dr = \mu \int_0^R r^2 m(r)\phi(r)\,dr.$$

Thus $(r^2m')' + \mu r^2 m = 0$ distributionally on $(0,R)$, and ordinary one-dimensional regularity makes m smooth there. Putting $k = \sqrt{\mu}$, the general solution is $(A\sin(kr) + B\cos(kr))/r$. Continuity of g at q gives $m(r) \to g(q)$ as $r \downarrow 0$, so $B = 0$ and $A = g(q)/k$. Consequently

$$m(r) = g(q)\frac{\sin(kr)}{kr}.$$

Since $|V_R| = |\Sigma|R^3/3$, integration gives

$$\frac{1}{|V_R|}\int_{V_R} g = \frac{3}{R^3}\int_0^R r^2 m(r)\,dr = \frac{3(\sin(kR) - kR\cos(kR))}{(kR)^3}\,g(q),$$

which proves (6.1). For $0 < kR < \pi/2$, the integral representation shows $A_0(kR) > 0$. If $g > 0$ in V_R, its integral is positive, and (6.1) gives $g(q) > 0$. □

Proposition 6.3 (Strict segment monotonicity up to the boundary)**.** *Let $P \subset \mathbb{R}^3$ be a bounded convex polytope, and let $u \in H^2(P)$ be a weak Neumann eigenfunction with eigenvalue $\mu > 0$. Suppose $p_0, p_1 \in \overline{P}$, $p_0 \neq p_1$, and put $w = p_1 - p_0$. If*

$$D_w u > 0 \qquad \textit{throughout } P, \tag{6.2}$$

then $u(p_1) > u(p_0)$. Reversing both inequalities gives the corresponding decreasing statement.

Proof. Set $g = D_w u \in H^1(P)$ and $q = (p_0 + p_1)/2$. Every face incident at q contains both p_0 and p_1: its defining affine function is nonnegative at both endpoints and zero at their midpoint, hence zero at both endpoints. Thus w is tangent to every incident face plane. This condition is vacuous when q is interior.

Choose $\rho > 0$ so that $B_\rho(q)$ meets no face plane not containing q. As in the cone-mean proof, P agrees with its incident cone in this ball, and translations by tw preserve all the incident half-spaces. Put $U = P \cap B_{\rho/2}(q)$. For $0 < |t||w| < \rho/4$, the difference quotient

$$g_t(x) = \frac{u(x+tw) - u(x)}{t}$$

is therefore well defined on U. Given an $H^1(U)$ test function η supported away from the spherical boundary, translate it to $\eta(\,\cdot - tw)$ on the shifted neighborhood and set it to zero elsewhere in P. This is an admissible $H^1(P)$ test: the incident half-spaces are invariant and the shifted support stays inside $B_\rho(q)$. Changing variables in the weak equation for u, subtracting its unshifted version and dividing by t gives

$$\int_U \nabla g_t \cdot \nabla\eta = \mu \int_U g_t \eta.$$

Since P is Lipschitz, take an $H^2(\mathbb{R}^3)$ extension $\widehat{u}$ of u and put $G = D_w\widehat{u} \in H^1(\mathbb{R}^3)$. All points $x + stw$, $x \in U$ and $0 \le s \le 1$, remain in P. The Sobolev difference-quotient identity gives, in $H^1(U)$,

$$g_t = \int_0^1 G(\,\cdot + stw)\,ds.$$

Continuity of translations in $H^1(\mathbb{R}^3)$ then yields

$$\|g_t - g\|_{H^1(U)} \le \int_0^1 \|G(\,\cdot + stw) - G\|_{H^1(\mathbb{R}^3)}\,ds \longrightarrow 0.$$

Passing to the limit in the weak equation for g_t proves the local weak Neumann equation for g on every incident face simultaneously.

Apply Lemma 6.1 with radius $\rho/2$. For

$$0 < R < \min\{\rho/4, \pi/(2\sqrt{\mu})\},$$

positivity of g in P and Lemma 6.2 give $g(q) > 0$. Hence there are $r > 0$ and $\eta_0 > 0$ such that the continuous representative satisfies $g \geq \eta_0$ on $\overline{P} \cap B_r(q)$.

To compare endpoint values, fix $z \in P$ and, for $0 < \varepsilon < 1$, put

$$p_{i,\varepsilon} = (1-\varepsilon)p_i + \varepsilon z \quad (i = 0, 1).$$

Convexity puts the whole joining segment in P. Interior regularity of u permits integration along that segment:

$$u(p_{1,\varepsilon}) - u(p_{0,\varepsilon}) = (1-\varepsilon)\int_0^1 g((1-\varepsilon)((1-s)p_0 + sp_1) + \varepsilon z)\, ds.$$

Choose $0 < \delta < 1/2$ with $\delta|w| < r/2$. For sufficiently small ε, one also has $\varepsilon|z - q| < r/2$. The argument of g then lies in $B_r(q)$ whenever $|s - 1/2| < \delta$. The integrand is positive everywhere and at least η_0 on this fixed interval, so the right side is at least $2\delta\eta_0(1-\varepsilon)$. Continuity of u on $\overline{P}$ allows $\varepsilon \downarrow 0$ and gives $u(p_1) - u(p_0) \geq 2\delta\eta_0 > 0$. □

7. Canonical extrema and the complete classification

Proposition 7.1 (Exact extrema of a vertex mode)**.** *For every $j \in \{1, 2, 3, 4\}$,*

$$\arg\max_{\overline{T}} f_{v_j} = \{v_j\}, \qquad \arg\min_{\overline{T}} f_{v_j} = \{v_i : i \neq j\}. \tag{7.1}$$

There is a number $\alpha > 0$, independent of j, such that

$$f_{v_j}(v_i) = \begin{cases} \alpha, & i = j, \\ -\alpha/3, & i \neq j. \end{cases} \tag{7.2}$$

Proof. Use the coordinates of Lemma 5.1, with $j = 4$, and abbreviate $f = f_{v_4}$. Every point of $\overline{T}$ can be moved into $\overline{C}$ by a symmetry fixing v_4, and f is invariant under that symmetry. The restriction of f to C belongs to $H^2(C)$, and Lemma 5.1 gives zero normal trace on every chamber face, including the mirrors. Green's identity and $-\Delta f = \mu f$ therefore give the full weak Neumann equation on C. Thus Proposition 6.3 applies on this convex polytope.

Let $p = (r, t, z) \in \overline{C} \setminus \{v_4\}$. Since $r, t \geq 0$ and $z \leq 1$, with $z = 1$ possible only at v_4, one has $1 - z > 0$. The constant vector

$$w = v_4 - p = (-r, -t, 1 - z)$$

satisfies

$$D_w f = -rf_r - tf_t + (1-z)f_z > 0 \qquad \text{throughout } C.$$

Here r, t, z in the coefficients are the fixed coordinates of p; the derivatives are evaluated at an arbitrary point of C. The strict inequality follows from $1 - z > 0$ and $f_z > 0$. Proposition 6.3, applied to the segment from p to v_4 in $\overline{C}$, gives $f(p) < f(v_4)$. Symmetry reduction to $\overline{C}$ therefore proves that v_4 is the unique global maximum.

For the minimum, now take $p = (r, t, z) \in \overline{C} \setminus \{v_1\}$ and write $\lambda_i = \lambda_i(p)$. The barycentric formulas give

$$\frac{R}{2} - r = \frac{R}{2}(\lambda_4 + 3\lambda_3) \geq 0, \qquad \frac{\sqrt{3}R}{2} - t = \frac{\sqrt{3}R}{2}(2\lambda_2 + \lambda_3 + \lambda_4) \geq 0, \qquad -\frac{1}{3} - z = -\frac{4}{3}\lambda_4 \leq 0. \tag{7.3}$$

The target vertex v_1 lies in $\overline{C}$, and at least one of the three displayed coefficients is nonzero because $p \neq v_1$. The strict signs $f_r < 0$, $f_t < 0$, $f_z > 0$ on C therefore imply

$$D_{v_1 - p} f < 0 \qquad \text{throughout } C.$$

The decreasing version of Proposition 6.3 yields $f(p) > f(v_1)$. Hence v_1 is the unique minimum on $\overline{C}$. Its orbit under the symmetries fixing v_4 is $\{v_1, v_2, v_3\}$, so these, and only these, minimize f on $\overline{T}$. Relabeling gives (7.1) for all j.

The maximum $\alpha = f_{v_j}(v_j)$ is independent of j by equivariance. It is positive because f_{v_j} is nonzero and has mean zero. Equivariance also gives one common value β for every $f_{v_j}(v_i)$ with $i \neq j$: vertex permutations act transitively on these ordered pairs of distinct indices. Linearity and $\sum_j v_j = 0$ give $\sum_j f_{v_j} = 0$. Evaluating this identity at a vertex gives $\alpha + 3\beta = 0$, proving (7.2). □

Proposition 7.2 (Exact extrema of an opposite-edge-pair mode)**.** *For distinct indices i, j,*

$$\arg\max_{\overline{T}} f_{v_i+v_j} = \{v_i, v_j\}, \qquad \arg\min_{\overline{T}} f_{v_i+v_j} = \{v_k : k \notin \{i, j\}\}. \tag{7.4}$$

Its maximum and minimum values are $2\alpha/3$ and $-2\alpha/3$, respectively.

Proof. Take $i = 1, j = 2$ and the coordinates of Lemma 4.1. Write $h = f_{v_1+v_2}$. Reflections in $x = 0$ and $y = 0$ leave h invariant and move any point into the closed quarter chamber

$$\overline{Q} = \overline{T} \cap \{x \geq 0, y \geq 0\}.$$

The function $h|_Q$ is a weak Neumann eigenfunction on this convex polytope by the same Green's identity argument: restriction preserves H^2, and Lemma 4.1 supplies zero normal traces on its physical faces and mirrors.

For $p = (x, y, z) \in \overline{Q} \setminus \{v_1\}$, put

$$w = v_1 - p = (a - x, -y, b - z).$$

The defining inequalities (4.4) imply $y \geq 0$, $z \leq b$ and $0 \leq x \leq (z + b)/\sqrt{2} \leq \sqrt{2}b = a$. Thus all three terms in

$$D_w h = (a - x)h_x - yh_y + (b - z)h_z$$

are nonnegative on Q. If $z < b$, the last is strictly positive. If $z = b$, the upper face inequality forces $y = 0$, and $p \neq v_1$ forces $x < a$; then the first term is strictly positive because $h_x > 0$ on Q. In every case $D_w h > 0$ throughout Q. Proposition 6.3 gives $h(p) < h(v_1)$. Undoing the two reflections shows that the only global maxima are the orbit $\{v_1, v_2\}$ of v_1.

The orthogonal symmetry $(x, y, z) \mapsto (y, x, -z)$ interchanges the two vertex pairs and sends $v_1 + v_2$ to its negative. Equivariance gives $h(y, x, -z) = -h(x, y, z)$, so the minima are exactly v_3, v_4. Since $h = f_{v_1} + f_{v_2}$ by linearity, (7.2) gives the extreme values $\alpha - \alpha/3 = 2\alpha/3$ and $-2\alpha/3$. Vertex permutations prove the assertion for every pair. □

Proof of Theorem 1.1. The dimension and the linear family f_a are supplied by Lemma 3.1. Put

$$\kappa = \frac{4\alpha}{3} > 0, \qquad \psi_i = \kappa^{-1} f_{v_i}. \tag{7.5}$$

Equation (7.2) gives $\psi_i(v_j) = \delta_{ij} - 1/4$, and $\sum_i \psi_i = 0$. The spanning property of the vertex vectors and Lemma 3.1 show that the ψ_i span $E_1(T)$. Given any representation $u = \sum_i d_i \psi_i$, replace d_i by $c_i = d_i - \frac{1}{4}\sum_j d_j$. Since $\sum_i \psi_i = 0$, this leaves u unchanged and gives $\sum_i c_i = 0$. Evaluating at v_j now gives

$$u_c(v_j) = \sum_i c_i(\delta_{ij} - 1/4) = c_j.$$

Thus the zero-sum coefficients are uniquely determined by u, and every zero-sum vector c occurs as its vertex-value vector. Vertex evaluation is therefore an isomorphism. In particular, it proves the asserted uniqueness of each ψ_i from its prescribed vertex values. Equivariance gives the permutation rule for this family.

To classify extrema, take $c \neq 0$ and relabel the vertices so that

$$c_1 \geq c_2 \geq c_3 \geq c_4,$$

and set

$$A = c_1 - c_2, \qquad B = c_2 - c_3, \qquad C = c_3 - c_4.$$

These numbers are nonnegative, and at least one is positive because a constant zero-sum vector is zero. Subtracting c_3 from each coefficient leaves u_c unchanged, since $\sum_i \psi_i = 0$, and gives

$$u_c = A\psi_1 + B(\psi_1 + \psi_2) + C(-\psi_4). \tag{7.6}$$

The three displayed modes attain their global maxima at v_1. Their exact maximizing sets, by the two preceding propositions, are respectively

$$\{v_1\}, \qquad \{v_1, v_2\}, \qquad \{v_1, v_2, v_3\}. \tag{7.7}$$

Because the coefficients in (7.6) are nonnegative, $u_c(x) \le u_c(v_1) = c_1$ for every $x \in \overline{T}$. Equality holds exactly when x belongs to the maximizing set of every mode whose coefficient is positive: each summand has a nonnegative deficit from its value at v_1, and a sum of those deficits is zero precisely when each active deficit is zero.

If $A > 0$, the intersection is $\{v_1\}$; if $A = 0$ and $B > 0$, it is $\{v_1, v_2\}$; if $A = B = 0$, then $C > 0$ and the intersection is $\{v_1, v_2, v_3\}$. These are exactly the vertices for which $c_i = c_1$. This proves (1.5), with all ties included. Apply the same conclusion to $-u_c$ to obtain (1.6). The strict nonvertex statement follows at once. □

Corollary 7.3 (Norm, multiplicities of extrema and changes of scale)**.** *For every $c \in \mathbb{R}^4$ with $\sum_i c_i = 0$,*

$$\|u_c\|_{L^\infty(T)} = \max_i |c_i|.$$

For $u_c \neq 0$, the number of global maxima is the multiplicity of the largest vertex value, and the number of global minima is the multiplicity of the smallest vertex value. If $\widetilde{T} = x_0 + sRT$, where $s > 0$ and R is orthogonal, the corresponding eigenfunctions are $\widetilde{u}(x) = u(R^{-1}(x - x_0)/s)$, the eigenvalue is $s^{-2}\mu_1(T)$, and the extrema in the theorem are carried to $x_0 + sRv_i$.

Proof. For $c = 0$ the norm identity is immediate. Otherwise, (1.5)–(1.6) give $\max_{\overline{T}} |u_c| = \max_i |c_i|$ and the stated counts. Continuity on $\overline{T}$ identifies this maximum with the essential supremum on T: every sufficiently small neighborhood of a maximizing point meets T in a nonempty open set, of positive measure, on which $|u_c|$ is arbitrarily close to its maximum.

For the similarity $F(y) = x_0 + sRy$, the pullback $\widetilde{u} = u \circ F^{-1}$ bijects the mean-zero H^1 spaces on T and $\widetilde{T}$. Change of variables gives, for every nonzero $u \in H^1(T)$,

$$\frac{\int_{\widetilde{T}} |\nabla \widetilde{u}|^2}{\int_{\widetilde{T}} |\widetilde{u}|^2} = s^{-2} \frac{\int_T |\nabla u|^2}{\int_T |u|^2}.$$

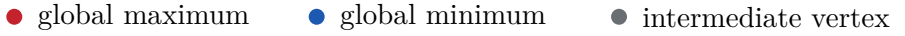
The variational characterization of the first positive eigenvalue therefore gives $\mu_1(\widetilde{T}) = s^{-2}\mu_1(T)$. The weak eigenvalue equation transforms by the same change of variables, giving the stated correspondence of first eigenspaces. Finally, F is a bijection of the closures and $\widetilde{u}(F(y)) = u(y)$, so it carries each exact extremal set to its claimed image. □

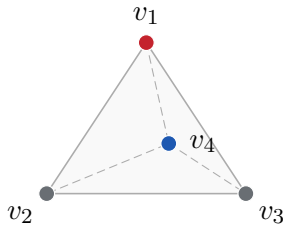
8. Pictures of the extremal sets

The coordinates in the following pictures are schematic projections of the solid tetrahedron. Each displayed coefficient vector is nonzero and has zero sum, so Theorem 1.1 supplies a unique eigenfunction u_c with these actual vertex values. A red vertex is a global maximum, a blue vertex is a global minimum, and a gray vertex has an intermediate value. Dashed lines indicate projected hidden edges. By the same theorem, every nonvertex point, including points of an edge joining two equally colored vertices, has value strictly between the global minimum and maximum.

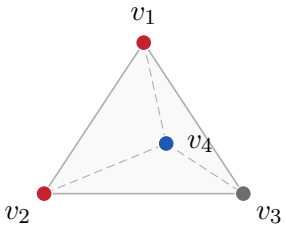
If m, n are the numbers of global maxima and minima of a nonzero first eigenfunction, their vertex sets are nonempty and disjoint, so $m, n \geq 1$ and $m + n \leq 4$. The four panels and their negatives realize all six possible pairs:

$$(m, n) \in \{(1,1), (2,1), (1,2), (2,2), (1,3), (3,1)\}.$$



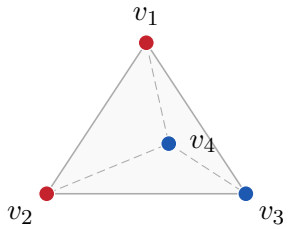


(a) $c = (3, 1, -1, -3)$.
One maximum and one minimum.

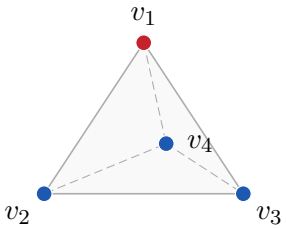


(b) $c = (2, 2, -1, -3)$.
Two maxima and one minimum.

FIGURE 1. Distinct extremes and a twofold maximum. In panel (b), no point of the open edge (v_1, v_2) is a global maximum.



(c) $c = (1, 1, -1, -1)$.
Two maxima and two minima.



(d) $c = (3, -1, -1, -1)$.
One maximum and three minima.

FIGURE 2. Panel (c) shows $u_c = 2(\psi_1 + \psi_2)$, a positive multiple of the pair mode; panel (d) shows $u_c = 4\psi_1$, a positive multiple of the vertex mode. In panel (d), the minimum occurs at the three blue vertices, and at no other point of their triangular face. Replacing u_c by $-u_c$ interchanges red and blue.

Remark 8.1 (Scope of the result). The theorem concerns the Euclidean Neumann Laplacian on the solid regular tetrahedron and the whole eigenspace of its first positive eigenvalue. The argument uses the full tetrahedral symmetry and the specific sparse normal relations of its stabilizer chambers. A precise way to see the effect of an affine deformation is to write $F(y) = x_0 + Ay$, with A invertible. If u is a Euclidean Neumann eigenfunction on $F(T)$ with eigenvalue λ, its pullback $v = u \circ F$ satisfies

$$\int_T B\nabla v \cdot \nabla\varphi = \lambda \int_T v\varphi \qquad (\varphi \in H^1(T)), \qquad B = A^{-1}A^{-\mathsf{T}}.$$

Indeed, this is the change of variables in the weak equation; the common Jacobian factor $|\det A|$ cancels. The resulting boundary condition is $(B\nabla v) \cdot \nu = 0$. The matrix B is scalar precisely when $A = sR$ for $s > 0$ and R orthogonal, the similarity case of Corollary 7.3. For general A, the chamber transformations and symmetry arguments above require additional justification. The variational equality used in the proof also concerns the first positive eigenspace. Extensions to arbitrary tetrahedra or to higher eigenvalues therefore require further arguments.

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